Symbiosis Institute of Technology, Pune
Bachelor of Technology (Robotics and Artificial Intelligence)
Programme Structure 2026-2030

<table>
<tr><td>1.</td><td>OBJECTIVE</td><td colspan="4">To generate competent manpower in the emerging areas of Robotics and Artificial intelligence To inculcate among the students an aptitude for engineering and research in Industry 4.0 for generation of better and smarter solutions to real world problems.</td></tr>
<tr><td>2.</td><td>DURATION (IN MONTHS)</td><td colspan="4">48 (Full Time)</td></tr>
<tr><td>3.</td><td>INTAKE</td><td colspan="4">60</td></tr>
<tr><td rowspan="4">4.</td><td rowspan="4">RESERVATION</td><td rowspan="2">I.Within the sanctioned intake</td><td>a) SC
(In Percentage)</td><td>b) ST
(In Percentage)</td><td>c) Persons with Disabilities
(In Percentage)</td></tr>
<tr><td>15</td><td>7.5</td><td>3</td></tr>
<tr><td rowspan="2">II.Over and above the sanctioned intake</td><td colspan="2">a) Kashmiri Migrants
(In Seats)</td><td colspan="2">b) International Students
(In Percentage)</td></tr>
<tr><td colspan="2">2</td><td colspan="2">15</td></tr>
<tr><td>5.</td><td>ELIGIBILITY</td><td colspan="4">Passed 10+2 examination with Physics, Chemistry and Mathematics as compulsory subjects. Obtained at least 45% marks (40% marks in case of candidates belonging to reserved category) in the above subjects taken together. OR Passed D.Voc. Stream in the same or allied sector. (The University will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to prepare Level playing field and desired learning outcomes of the programme).
B.Tech. : Lateral Entry Passed Minimum Three-years/ Two-year (Lateral Entry) Diploma examination with at least 45% marks (40% marks in case of candidates belonging to reserved category) in ANY branch of Engineering and Technology.
OR
Passed B.Sc. Degree from a recognized University as defined by UGC, with at least 45% marks (40% marks or equivalent grade for Scheduled Caste / Scheduled Tribes) and passed 10+2 examination with Mathematics as a subject.
OR
Passed B. Voc/3-year D.Voc. Stream in the same allied sector. (The Constituent will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to achieve desired learning outcomes of the</td></tr>
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		programme).			
6.	**SELECTION PROCEDURE**	Merit list by valid score of Symbiosis Entrance Test (SITEEE) or Joint Entrance Examination (JEE - Main) or Any State Government Engineering Entrance Examination.			
7.	**MEDIUM OF INSTRUCTION**	English			
8.	**PROGRAMME PATTERN**	Semester			
9.	**COURSE & SPECIALISATION**	Annexure A: Bachelor of Technology (Robotics and Artificial Intelligence) Students may pursue optional 'Honours' OR 'Minor' specialisation in one of the specialisation areas by completing additional 20 credits in Semester: 5, 6 and 7 as specified in Annexure B for Honours and Annexure C for Minor Specialisation in Specialisation by completing additional 18 credits in Semester: 3, 4, 5 and 6 as specified in Annexure C. Annexure B: Optional 'Honours' specialisation area 1. Aerial Robotics and Drone Technology Annexure C: Optional 'Minor' specialisation area 1. Computer Science Engineering Fundamentals 2. Quantum Technologies			
10.	**FEE**		**Academic Fee p.a**	**Institute Deposit**	**Total**

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	Indian Students (Amount in INR)		300000	20000	320000
	International Students	**NRI/ PIO/ OCI Category (Amount in US$)**	5875	275	6150
		Foreign National Category (Amount in US$)	1300	275	1575

11.	**ASSESSMENT**	The theory courses will have 40% Continuous Assessment and 60% End Semester [University] examination, Lab courses (Practical) will have 60% Continuous Assessment and 40% End Semester [University] examination; however, some courses (not more than 30% of the total programme credits) may have 100% Continuous Assessment.
12.	**STANDARD OF PASSING**	The assessment of the student for each examination is done, based on relative performance. Maximum Grade Point (GP) is 10 corresponding to O (Outstanding). For all courses, a student is required to pass both internal and external examination separately with a minimum Grade Point of 4 corresponding to Grade P. Students securing less than 40% absolute marks in each head of passing will be declared FAIL. The University awards a degree to the student who has achieved a minimum CGPA of 4 out of maximum of 10 CGPA for the programme.
13.	**AWARD OF DEGREE**	Bachelor of Technology (Robotics and Artificial Intelligence) OR Bachelor of Technology (Robotics and Artificial Intelligence) with Honours in Aerial Robotics and Drone Technology OR Bachelor of Technology (Robotics and Artificial Intelligence) with Minor in Computer Science Engineering Fundamentals / Quantum Technologies will be awarded at the end of semester 8 examination by taking into consideration the performance of all semester examinations after obtaining a minimum 4.00 CGPA out of 10 CGPA.
14.	**CLASSIFICATION OF CREDITS**	

Semester	Basic	Engineering	Professional	Professional	Humanities and	Multidisciplinary	Project/	Indian	Total	No. of	No. of Non-

	Sciences	Sciences	Core	Elective	Social Sciences including Management	Open Electives	Internship/ Seminar	Knowledge System	Credits	Mandatory Non-Credit Course/s	Credit Audit Course/s
Track 1											
1	4	8	7	0	1	0	0	0	20	0	As per the student's choice
2	7	2	7	0	2	0	0	2	20	1 *	
3	3	0	16	0	1	0	0	0	20	1 *	
4	3	0	10	3	3	3	0	0	22	1 *	
5	0	1	11	8	0	3	0	0	23	2 *	
6	0	2	9	3	0	3	2	0	19	1 *	
7	0	0	10	6	0	0	6	0	22	1 *	
8	0	0	0	0	0	0	14	0	14	0	
Total	**17**	**13**	**70**	**20**	**7**	**9**	**22**	**2**	**160**	**0**	
Track 2											
1	4	8	7	0	1	0	0	0	20	0	As per the student's choice
2	7	2	7	0	2	0	0	2	20	1 *	
3	3	0	16	0	1	0	0	0	20	1 *	
4	3	0	10	3	3	3	0	0	22	1 *	
5	0	1	11	8	0	3	0	0	23	2 *	
6	0	2	9	3	0	3	2	0	19	1 *	
7	0	0	6	0	0	0	16	0	22	1 *	
8	0	0	0	0	0	0	14	0	14	0	
Total	**17**	**13**	**66**	**14**	**7**	**9**	**32**	**2**	**160**	**0**	
Optional Additional Courses (Honours)											

5	0	0	6	0	0	0	0	0	6	0	
6	0	0	7	0	0	0	0	0	7	0	
7	0	0	0	0	0	0	7	0	7	0	
Total	**0**	**0**	**13**	**0**	**0**	**0**	**7**	**0**	**20**	**0**	
Optional Additional Courses (Minor)											
3	0	0	5	0	0	0	0	0	5	0	As per the student's choice
4	0	0	8	0	0	0	0	0	8	0	
5	0	0	3	0	0	0	0	0	3	0	
6	0	0	2	0	0	0	0	0	2	0	
Total	**0**	**0**	**18**	**0**	**0**	**0**	**0**	**0**	**18**	**0**	
Optional Additional Courses (Minor) - Quantum Technologies											
3	0	0	3	0	0	0	0	0	3	0	As per the student's choice
4	0	0	6	0	0	0	0	0	6	0	
5	0	0	6	0	0	0	0	0	6	0	
6	0	0	3	0	0	0	0	0	3	0	
Total	**0**	**0**	**18**	**0**	**0**	**0**	**0**	**0**	**18**	**0**	

* Satisfactory completion of non credit courses 'Health and Wellness', '*Vasudhaiva Kutumbakam*' and 'Career Essentials I to V' is mandatory for award of degree.

Additional Note: #Health and Wellness Module I and Module II will be conducted during the semesters mentioned in the programme structure. However, the course will be listed on the students' grade sheets as "Health and Wellness" in the semester in which the institute's course code is officially assigned.

This Programme Structure is aligned with the norms laid down by the University and is approved by the Academic Council and Board of Management.
Hereafter changes (if any) which conform to the policy on "Curriculum Development and Review" would be permissible, subject to revision of the Programme Structure, following the specified processes.

Dean - Academics

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Symbiosis Institute of Technology, Pune
Bachelor of Technology (Robotics and Artificial Intelligence)
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Annexure A

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Semester : 1													
Generic Core Courses													
TEE7237	0701340101	Calculus	BS		3	1	0	0	0	40	60	4	100
TEE7342	0701340102	Biophysics and Mechanics	ES		3	0	0	0	0	30	45	3	75
TEE7539	0701340103	Principles of Electrical and Electronics Engineering	ES		2	0	2	15	10	20	30	3	75
TM2278	0701340104	Introduction to Environment and Sustainability	ES		0	0	2	25	0	0	0	1	25
TEE7364	0701340105	Tinker and IDEA Lab	ES		0	0	2	25	0	0	0	1	25
TEE7357	0701340106	Programming for Engineers	PC		2	0	2	15	10	20	30	3	75
TEE7538	0701340107	Introduction to Robotics	PC		2	0	2	15	10	20	30	3	75
TEE7537	0701340108	Fundamentals of AI	PC		0	0	2	15	10	0	0	1	25
T6873	0701340109	Creative Thinking	HSMC		1	0	0	0	0	25	0	1	25
				Total	**13**	**1**	**12**	**110**	**40**	**155**	**195**	**20**	**500**
Semester : 2													
Generic Core Courses													
TEE7257	0701340201	Statistics for Data Science	BS		3	1	0	0	0	40	60	4	100
TEE7255	0701340202	Linear Algebra	BS		2	1	0	0	0	30	45	3	75
TEE7540	0701340203	Python Essentials	ES		0	0	4	30	20	0	0	2	50
TEE7034	0701340204	Data Preprocessing and EDA Lab	PC		0	0	4	30	20	0	0	2	50

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
TEE7541	0701340205	Robotics Sensors and Actuators	PC		0	0	4	30	20	0	0	2	50
TEE7091	0701340206	Kinematics of Robotics	PC		2	0	2	25	0	50	0	3	75
T6732	0701340207	Critical Thinking	HSMC		1	0	0	0	0	25	0	1	25
THM6150	0701340208	Technical and Professional Communication Skills	HSMC		0	0	2	25	0	0	0	1	25
THM6144	0701340209	Indian Knowledge Systems	IKS		2	0	0	0	0	50	0	2	50
TEE7265	0701340210	Career Essentials - I *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
Total					**10**	**2**	**16**	**140**	**60**	**195**	**105**	**20**	**500**
Semester : 3													
Generic Core Courses													
TE7696	0701340301	Discrete Mathematics	BS		2	1	0	0	0	30	45	3	75
TEE7542	0701340302	Supervised and Unsupervised Machine Learning	ES		2	0	4	30	20	20	30	4	100
TEE7497	0701340303	Robotic Control Systems	PC		2	0	2	15	10	20	30	3	75
TEE7490	0701340304	Hydraulic and Pneumatic Systems	PC		2	0	2	15	10	20	30	3	75
TEE7536	0701340305	Data Structures and Algorithmic Techniques	PC		2	0	2	15	10	20	30	3	75

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
TEE7407	0701340306	Mechatronics	PC		1	0	0	0	0	25	0	1	25
TEE7492	0701340307	Mechatronics Lab	PC		0	0	2	15	10	0	0	1	25
TEE7401	0701340308	Total Productive Maintenance	PC		0	0	2	15	10	0	0	1	25
THM6277	0701340309	Principles of Economics	HSMC		1	0	0	0	0	25	0	1	25
TEE7419	0701340310	Career Essentials - II *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
TH4788		Health and Wellness Module I #	MC		0	0	0	0	0	0	0	0	0
				Total	**12**	**1**	**14**	**105**	**70**	**160**	**165**	**20**	**500**
Semester : 4													
Generic Core Courses													
TE7699	0701340401	Probability and Random Processes	BS		2	1	0	0	0	30	45	3	75
TEE7496	0701340402	Programmable Logic Controller and Human Machine Interface	PC		2	0	4	30	20	20	30	4	100
TEE7495	0701340403	Process Control and Instrumentation	PC		2	0	4	30	20	20	30	4	100
TEE7397	0701340404	Servo Motors and Drives	PC		1	0	0	0	0	25	0	1	25
TEE7398	0701340405	Servo Motors and Drives Lab	PC		0	0	2	15	10	0	0	1	25
P4618	0701340406	Service Learning	HSMC		0	0	4	50	0	0	0	2	50

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
T2646	0701340407	Entrepreneurship Venture	HSMC		1	0	0	0	0	25	0	1	25
TEE7420	0701340408	Career Essentials - III *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
TH4789		Health and Wellness Module II #	MC		0	0	0	0	0	0	0	0	0
Total					**8**	**1**	**14**	**125**	**50**	**120**	**105**	**16**	**400**
Professional Elective (Choose any one Course)													
TEE7498	0701340409	Robotic Perception and Actuation	PE		2	0	2	15	10	20	30	3	75
P6134	0701340410	Cloud Computing Essentials	PE		2	0	2	15	10	20	30	3	75
TEE7504	0701340411	Unmanned Aerial Vehicles	PE		2	0	2	15	10	20	30	3	75
Total Required Credits								**15**	**10**	**20**	**30**	**3**	**75**
Multidisciplinary Open Elective Courses (Choose any one Course)													
TEE7416	0701340412	Quantum Computing for Engineers	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
TEE7414	0701340413	Mathematics for Data Science	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
TEE7438	0701340414	Smart Cities Planning and Management	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75
TEE7435	0701340415	Intelligent Waste Management Techniques	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
TEE7458	0701340416	Web Technologies	MOPE	Computer Science and Engineering	2	1	0	0	0	30	45	3	75
TEE7447	0701340417	Data Science	MOPE	Computer Science and Engineering	2	1	0	0	0	30	45	3	75
TEE7018	0701340418	Engineering Simulation and Modeling Tools	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75
TEE7472	0701340419	Medical Electronics	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75
TE7351	0701340420	3D Printing and Prototyping	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
TEE7476	0701340421	Battery Management Systems	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
TEE7425	0701340422	Fundamentals of Machine Learning	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
TEE7418	0701340423	AI System Development	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
Total Required Credits								**0**	**0**	**30**	**45**	**3**	**75**
Semester : 5													
Generic Core Courses													
TEE7478	0701340501	Design Thinking	ES		1	0	0	0	0	25	0	1	25
P5740	0701340502	Manufacturing Systems	PC		2	0	4	30	20	20	30	4	100

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
P4361	0701340503	Microcontroller and Embedded Systems	PC		2	0	2	15	10	20	30	3	75
P6135	0701340504	Deep Learning	PC		2	0	2	15	10	20	30	3	75
TEE7385	0701340505	Applied ARVR	PC		0	0	2	15	10	0	0	1	25
P4784	0701340506	Career Essentials - IV *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
SMC001	0701340507	*Vasudhaiva Kutumbakam* *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
				Total	**7**	**0**	**10**	**75**	**50**	**85**	**90**	**12**	**300**
Professional Elective (Choose any one Group) **Professional Elective Group -1**													
P5738	0701340508	Robotics	PE		2	0	4	30	20	20	30	4	100
TEE7486	0701340509	Automation	PE		1	0	0	0	0	25	0	1	25
TEE7391	0701340510	Automation Lab	PE		0	0	4	30	20	0	0	2	50
TEE7502	0701340511	Simulation and Programming Lab	PE		0	0	2	15	10	0	0	1	25
				Total Required Credits				**75**	**50**	**45**	**30**	**8**	**200**

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Professional Elective Group - 2													
P5741	0701340512	Industrial Internet of Things	PE		2	0	4	30	20	20	30	4	100
P5742	0701340513	AI for Manufacturing	PE		2	0	4	30	20	20	30	4	100
Total Required Credits								**60**	**40**	**40**	**60**	**8**	**200**
Multidisciplinary Open Elective Courses (Choose any one Course)													
P5210	0701340514	Financial Mathematics	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
P5211	0701340515	Advanced Materials	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
P4656	0701340516	Sustainability Engineering- Design and Innovation	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75
P4658	0701340517	Occupational Health and Safety Management	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75
P4840	0701340518	Introduction to Cloud Computing	MOPE	Computer Science and Engineering	2	1	0	0	0	30	45	3	75
P4844	0701340519	Agile Methodologies	MOPE	Computer Science and Engineering	2	1	0	0	0	30	45	3	75
P4472	0701340520	Embedded System and IoT	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75
P5241	0701340521	Basics of 5G Technology	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
TE7989	0701340522	Electric and Hybrid Vehicles	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
T7650	0701340523	Six Sigma	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
P4789	0701340524	Optimization for ML Systems	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
P4790	0701340525	Deep Learning Essentials	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
Total Required Credits								**0**	**0**	**30**	**45**	**3**	**75**
Semester : 6													
Generic Core Courses													
P6987	0701340601	Natural Language Processing For Robotics	ES		0	0	4	30	20	0	0	2	50
P5782	0701340602	Robot Operating System	PC		2	0	4	30	20	20	30	4	100
P5225	0701340603	Computer Vision and Image Processing	PC		2	0	2	15	10	20	30	3	75
F0001	0701340604	Flexi-Credit Course	PC		1	0	0	0	0	25	0	1	25
TE7876	0701340605	Responsible AI for Robotics	PC		1	0	0	0	0	25	0	1	25
P5730	0701340606	Advanced Microcontrollers and Microprocessors	PC		2	0	2	15	10	20	30	3	75
P5783	0701340607	Project Based Learning	PIS		0	0	4	30	20	0	0	2	50

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
P4785	0701340608	Career Essentials - V *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
Total					**8**	**0**	**16**	**120**	**80**	**110**	**90**	**16**	**400**
Multidisciplinary Open Elective Courses (Choose any one Course)													
P5212	0701340609	Bioinformatics	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
P5213	0701340610	Introduction to Space Science	MOPE	Applied Science	2	1	0	0	0	30	45	3	75
P4657	0701340611	GIS and Remote Sensing Analytics	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75
P4659	0701340612	Environmental Impact Assessment	MOPE	Civil Engineering	2	1	0	0	0	30	45	3	75
P4830	0701340613	Software Testing and Quality Assurance	MOPE	Computer Science and Engineering and Information Technology	2	1	0	0	0	30	45	3	75
P4831	0701340614	Introduction to AR-VR	MOPE	Computer Science and Engineering and Information Technology	2	1	0	0	0	30	45	3	75
P4475	0701340615	Renewable Energy Systems	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75

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								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
P4274	0701340616	Semiconductor Technology Trends	MOPE	Electronics and Telecommunication Engineering	2	1	0	0	0	30	45	3	75
P4448	0701340617	Supply Chain Management	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
P4449	0701340618	Smart Manufacturing and Introduction of Industry 5.0	MOPE	Mechanical Engineering	2	1	0	0	0	30	45	3	75
P4792	0701340619	Data Engineering and Applications	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
P4791	0701340620	Gen AI Tools and Techniques	MOPE	Artificial Intelligence and Machine Learning	2	1	0	0	0	30	45	3	75
Total Required Credits								**0**	**0**	**30**	**45**	**3**	**75**
Semester : 7													
Generic Core Courses - Track 1													
F0004	0701340701	Flexi-Credit Course	PC		3	0	2	25	0	75	0	4	100
P4534	0701340702	Reinforcement Learning	PC		2	0	2	15	10	20	30	3	75
P5746	0701340703	Swarm Robotics	PC		2	0	2	15	10	20	30	3	75
F0002	0701340704	Flexi-Credit Course	PIS		0	0	4	50	0	0	0	2	50
T7804	0701340705	Project	PIS		0	0	8	60	40	0	0	4	100

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Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
SMC003	0701340706	Health and Wellness *	MC		0	0	0	0	0	0	0	Mandatory Non-Credit Course	0
Total Required Credits								**165**	**60**	**115**	**60**	**16**	**400**
Professional Elective-I (Choose any one Course)													
P5743	0701340707	Embedded Machine Learning	PE		2	1	0	0	0	30	45	3	75
P4536	0701340708	Foundations of GEN AI and LLM	PE		2	1	0	0	0	30	45	3	75
Total Required Credits								**0**	**0**	**30**	**45**	**3**	**75**
Professional Elective-II (Choose any one Course)													
P4897	0701340709	Hardware and Network Security	PE		2	1	0	0	0	30	45	3	75
P4786	0701340710	Graph Neural Networks and Applications	PE		2	1	0	0	0	30	45	3	75
Total Required Credits								**0**	**0**	**30**	**45**	**3**	**75**
Generic Core Courses - Track 2													
T7804	0701340705	Project	PIS		0	0	8	60	40	0	0	4	100
F0003	0701340711	Flexi-Credit Course	PC		3	0	0	0	0	75	0	3	75
F0003	0701340712	Flexi-Credit Course	PC		3	0	0	0	0	75	0	3	75

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Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Total Required Credits								**60**	**40**	**150**	**0**	**10**	**250**
Generic Elective courses Group - (Choose any one Course)													
T7912	0701340713	Startup Internship	PIS		0	0	24	180	120	0	0	12	300
T7912	0701340714	Research Internship	PIS		0	0	24	180	120	0	0	12	300
T7912	0701340715	Industry Internship	PIS		0	0	24	180	120	0	0	12	300
Total Required Credits								**180**	**120**	**0**	**0**	**12**	**300**
Semester : 8													
Generic Core Courses													
T7912	0701340801	Internship	PIS		0	0	24	180	120	0	0	12	300
T7802	0701340802	Seminar	PIS		0	0	4	30	20	0	0	2	50
Total					**0**	**0**	**28**	**210**	**140**	**0**	**0**	**14**	**350**

Abbreviations (Nature)	Description
BS	Basic Sciences
ES	Engineering Sciences
PC	Professional Core
PE	Professional Elective
HSMC	Humanities and Social Sciences including Management
MOPE	Multidisciplinary Open Electives
PIS	Project, Internship, Seminar
IKS	Indian Knowledge System
L	Lecture
MC	Mandatory Course
T	Tutorial
CA	Continuous Assessment
ESE	End Semester Examination
LAB	Laboratory

Track 1 (T1): For Regular Students
Track 2 (T2): For Students opting for Internship/ Entrepreneurship

Definition:
Honours: Students have the option to pursue an "Honours" degree by completing an additional 20 credits within their major discipline, focusing on more advanced, specialised, emerging, or multidisciplinary courses beyond the standard requirements of the B.Tech degree.

Minors: Students have the option to pursue a "Minor" by completing 18 credits in a discipline/ specialisation other than their major discipline beyond the standard requirements of the B.Tech. Degree.

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Semester	Continuous Assessment	End Semester Examination	Total Credits	Total Marks
Track 1				
Semester 1	3	17	20	500
Semester 2	7	13	20	500
Semester 3	2	18	20	500
Semester 4	4	18	22	550
Semester 5	2	21	23	575
Semester 6	2	17	19	475
Semester 7	6	16	22	550
Semester 8	0	14	14	350
Total	**26**	**134**	**160**	**4000**
Track 2				
Semester 1	3	17	20	500
Semester 2	7	13	20	500
Semester 3	2	18	20	500
Semester 4	4	18	22	550
Semester 5	2	21	23	575
Semester 6	2	17	19	475
Semester 7	6	16	22	550
Semester 8	0	14	14	350
Total	**26**	**134**	**160**	**4000**

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Annexure B
Optional 'Honours' Specialisation

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Semester : 5													
Specialisation Core Courses : Aerial Robotics and Drone Technology													
TE7850	0701340526	Introduction to Aerial Robotics and Drones	PC	Aerial Robotics and Drone Technology	3	0	0	0	0	30	45	3	75
P5288	0701340527	Motion Planning and Control	PC	Aerial Robotics and Drone Technology	2	0	2	15	10	20	30	3	75
				Total	**5**	**0**	**2**	**15**	**10**	**50**	**75**	**6**	**150**
Semester : 6													
Specialisation Core Courses : Aerial Robotics and Drone Technology													
TE7884	0701340621	Robotics Mobility and Perception	PC	Aerial Robotics and Drone Technology	3	0	0	0	0	30	45	3	75
TE7883	0701340622	Robotics Estimation and Learning	PC	Aerial Robotics and Drone Technology	2	0	0	0	0	20	30	2	50
TE7866	0701340623	Navigation and Communication Lab	PC	Aerial Robotics and Drone Technology	2	0	4	30	20	0	0	2	50
				Total	**7**	**0**	**4**	**30**	**20**	**50**	**75**	**7**	**175**
Semester : 7													
Specialisation Core Courses : Aerial Robotics and Drone Technology													

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Annexure B
Optional 'Honours' Specialisation

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
T7802	0701340717	Honours Seminar	PIS	Aerial Robotics and Drone Technology	0	0	4	30	20	0	0	2	50
T7805	0701340716	Honours Project	PIS	Aerial Robotics and Drone Technology	0	0	10	0	0	75	50	5	125
				Total	**0**	**0**	**14**	**30**	**20**	**75**	**50**	**7**	**175**

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Annexure B
Optional 'Honours' Specialisation

Semester	Continuous Assessment	End Semester Examination	Total Credits	Total Marks
Aerial Robotics and Drone Technology				
Semester 5	0	6	6	150
Semester 6	0	7	7	175
Semester 7	0	7	7	175
Total	**0**	**20**	**20**	**500**

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Annexure C
Optional 'Minor' Specialisation

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Semester : 3													
Specialisation Core Courses : Computer Science Engineering Fundamentals													
TEE7290	0701340311	Computer Architecture and Organization	PC	Computer Science Engineering Fundamentals	2	0	2	30	20	10	15	3	75
TEE7456	0701340312	Programming with JAVA	PC	Computer Science Engineering Fundamentals	0	0	4	30	20	0	0	2	50
				Total	**2**	**0**	**6**	**60**	**40**	**10**	**15**	**5**	**125**
Semester : 3													
Specialisation Core Courses : Quantum Technologies													
TEE7550	0701340313	Foundations of Quantum Computing	PC	Quantum Technologies	3	0	0	0	0	30	45	3	75
				Total	**3**	**0**	**0**	**0**	**0**	**30**	**45**	**3**	**75**
Semester : 4													
Specialisation Core Courses : Computer Science Engineering Fundamentals													
TEE7442	0701340424	Computer Networks	PC	Computer Science Engineering Fundamentals	3	0	2	15	10	30	45	4	100
TEE7449	0701340425	Database Management Systems	PC	Computer Science Engineering Fundamentals	2	0	4	30	20	20	30	4	100
				Total	**5**	**0**	**6**	**45**	**30**	**50**	**75**	**8**	**200**
Semester : 4													

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Annexure C
Optional 'Minor' Specialisation

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
Specialisation Core Courses : Quantum Technologies													
TEE7551	0701340426	Foundations of Quantum Technologies	PC	Quantum Technologies	3	0	0	0	0	30	45	3	75
TEE7554	0701340427	Quantum Technologies Lab	PC	Quantum Technologies	2	0	2	15	10	30	20	3	75
		Total			**5**	**0**	**2**	**15**	**10**	**60**	**65**	**6**	**150**
Semester : 5													
Specialisation Core Courses : Computer Science Engineering Fundamentals													
P5300	0701340528	Operating Systems	PC	Computer Science Engineering Fundamentals	3	0	0	0	0	75	0	3	75
		Total			**3**	**0**	**0**	**0**	**0**	**75**	**0**	**3**	**75**
Semester : 5													
Specialisation Core Courses : Quantum Technologies													
TEE7553	0701340529	Introduction to Quantum Computation	PC	Quantum Technologies	3	0	0	0	0	30	45	3	75
TEE7555	0701340530	Survey of Quantum Technologies and Applications	PC	Quantum Technologies	3	0	0	0	0	30	45	3	75
		Total			**6**	**0**	**0**	**0**	**0**	**60**	**90**	**6**	**150**
Semester : 6													
Specialisation Core Courses : Computer Science Engineering Fundamentals													

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Annexure C
Optional 'Minor' Specialisation

Catalog Course Code	Course Code	Course Title	Nature	Specialisation/ Area/ Department	Teaching Scheme (Hours Per Week)			Examination Scheme (Marks)				Total Credits	Total
								Practical		Theory			
					L	T	Lab	CA	ESE	CA	ESE		
T7674	0701340624	Cyber Security	PC	Computer Science Engineering Fundamentals	1	0	2	25	0	25	0	2	50
Total					**1**	**0**	**2**	**25**	**0**	**25**	**0**	**2**	**50**
Semester : 6													
Specialisation Core Courses : Quantum Technologies													
TEE7552	0701340625	Introduction to Quantum Communication	PC	Quantum Technologies	3	0	0	0	0	30	45	3	75
Total					**3**	**0**	**0**	**0**	**0**	**30**	**45**	**3**	**75**

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Annexure C
Optional 'Minor' Specialisation

Semester	Continuous Assessment	End Semester Examination	Total Credits	Total Marks
Computer Science Engineering Fundamentals				
Semester 3	0	5	5	125
Semester 4	0	8	8	200
Semester 5	3	0	3	75
Semester 6	2	0	2	50
Total	**5**	**13**	**18**	**450**
Quantum Technologies				
Semester 3	0	3	3	75
Semester 4	0	6	6	150
Semester 5	0	6	6	150
Semester 6	0	3	3	75
Total	**0**	**18**	**18**	**450**